



NORTH SYDNEY GIRLS HIGH SCHOOL

HSC Mathematics Extension 2

Assessment Task 1

Term 4 2012

Name: _____ **Mathematics Class:** _____

Student Number _____

Time Allowed: 50 minutes + 2 minutes reading time

Available Marks: 45

Instructions:

Question 1 (a) Multiple choice (10 marks)

- Indicate your answer by colouring the appropriate circle

(b) Constrained answer (marks)

- Indicate your answer by entering it into the diagram on the answer sheet provided

Question 2 Free response(10 marks)

- Write your answers on the examination booklet provided
- Write on one side of the page only
- Do not work in columns
- Attempt all questions
- Show all necessary working
- Marks may be deducted for incomplete or poorly arranged work

Question	1 -10	11a	11 bc	12	Total	
E3	/10	/5			/15	
E2				/10	/10	
E9			/		/	
					/	%

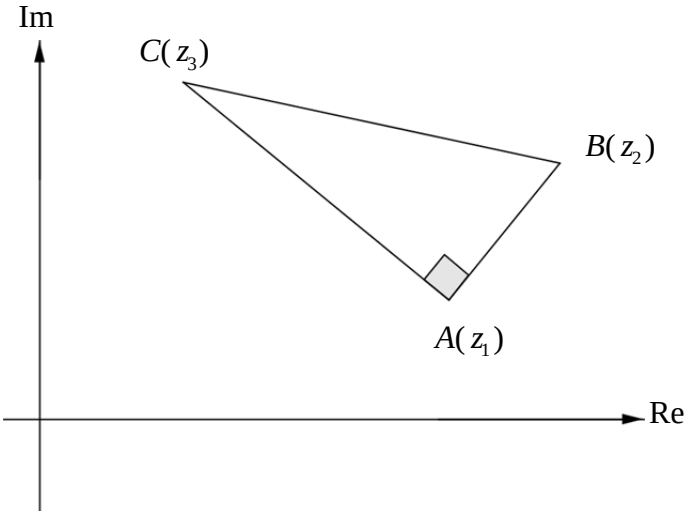
Question 1

Multiple Choice (10 marks). Please answer on the answer sheet provided.

- (1) Let $z = 5 - 2i$ and $w = 3 + 4i$. Then $z - \bar{w}$ is
- (A) $2 - 6i$ (B) $2 + 2i$ (C) $8 - 6i$ (D) $2 - 2i$
- (2) $\frac{1}{i^{101}} =$
- (A) 1 (B) -1 (C) i (D) $-i$
- (3) If $z = a + ib$, where $a \neq 0$ and $b \neq 0$, which of the following statements is false?
- (A) $z - \bar{z} = 2bi$ (B) $|z| + |\bar{z}| = |z + \bar{z}|$
- (C) $\arg(z) + \arg(\bar{z}) = 0$ (D) $|z|^2 = z\bar{z}$
- (4) If $z = i - \sqrt{3}$ then $\arg(z^4)$ is
- (A) $\frac{2\pi}{3}$ (B) $\frac{\pi}{3}$
- (C) $-\frac{2\pi}{3}$ (D) $-\frac{\pi}{3}$
- (5) The non-real roots of $z^3 + 8 = 0$ have
- (A) modulus 8 and arguments $\pm \frac{2\pi}{3}$ (B) modulus 8 and arguments $\pm \frac{\pi}{3}$
- (C) modulus 2 and arguments $\pm \frac{2\pi}{3}$ (D) modulus 2 and arguments $\pm \frac{\pi}{3}$
- (6) If $z = 2i$ and $w = 2 - 2i$, then the modulus and argument of $\frac{z}{w}$ are respectively
- (A) $\frac{1}{\sqrt{2}}, \frac{3\pi}{4}$ (B) $\frac{1}{\sqrt{2}}, -\frac{\pi}{4}$
- (C) $\sqrt{2}, \frac{3\pi}{4}$ (D) $\sqrt{2}, \frac{\pi}{4}$

- (7) Let $u = 4(\cos \frac{3\pi}{4} + i \sin \frac{3\pi}{4})$ and $v = r(\cos \theta + i \sin \theta)$.
If $uv = 12(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3})$, then
- (A) $r = 3, \theta = \frac{5\pi}{12}$ (B) $r = 8, \theta = -\frac{5\pi}{12}$
(C) $r = 8, \theta = \frac{5\pi}{12}$ (D) $r = 3, \theta = -\frac{5\pi}{12}$
- (8) The point R represents the complex number z on the Argand diagram. Which of the following describes the locus of R specified by $|z - 6| = |z|$?
- (A) Circle centre $(6, 0)$, radius $|z|$
(B) Circle centre the origin, radius $|z|$
(C) Perpendicular bisector of $(0, 0)$ and $(6, 0)$
(D) Perpendicular bisector of $(0, 0)$ and $(0, 6)$
- (9) The equation $x^2 - 2ix + 3 = 0$ has
- (A) no roots (B) one real and one complex root
(C) two purely imaginary roots (D) two real roots
- (10) The points A, B and C are three points on the Argand plane representing the complex numbers z_1, z_2 and z_3 respectively as shown on the diagram. AB is perpendicular to AC. Which statement is correct?

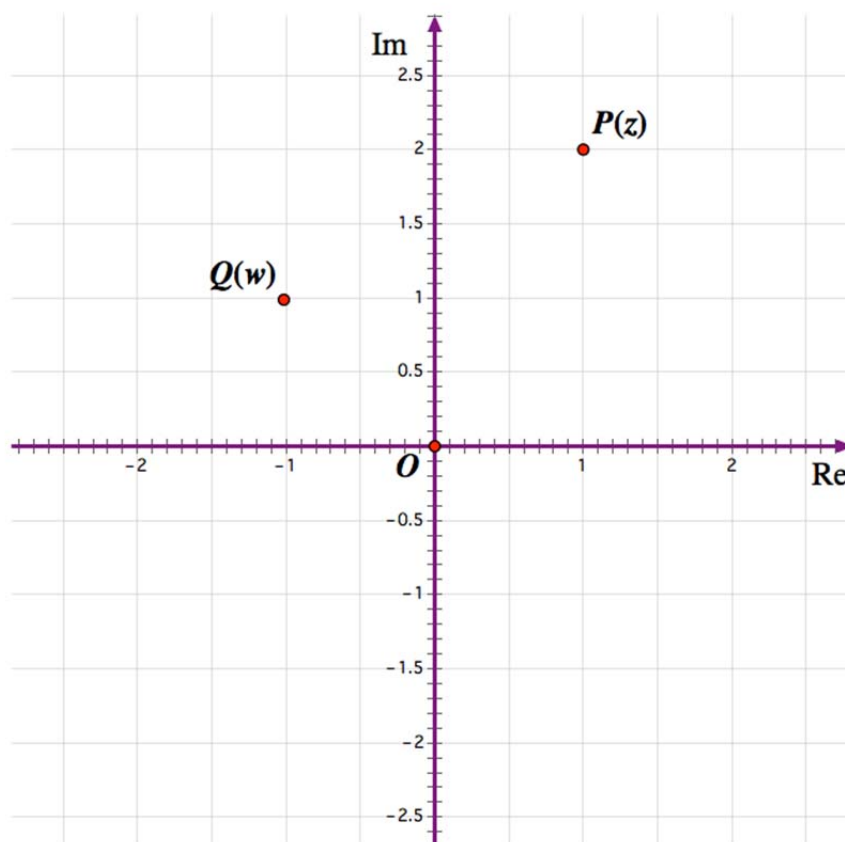
(A) $\arg\left(\frac{z_3 - z_2}{z_3 - z_1}\right) = \frac{\pi}{2}$
(B) $\arg\left(\frac{z_3 - z_1}{z_2 - z_1}\right) = \frac{\pi}{2}$
(C) $\arg[(z_3 - z_1)(z_2 - z_1)] = \frac{\pi}{2}$
(D) $\arg[(z_3 - z_2)(z_3 - z_1)] = \frac{\pi}{2}$



Question 11 continued: Constrained Answers (20 marks) .

Indicate your answer by entering it into the appropriate diagrams on the answer sheet provided.

(a) Point P represents the complex number z and Q represents the complex number w .



Plot the points that represent

5

- (i) $A(\bar{z})$
- (ii) $B(-w)$
- (iii) $C\left(\frac{1}{w}\right)$
- (iv) $D(iz)$
- (v) $E(w - z)$

(b)

- (i) Plot the roots of $z^8 - 1 = 0$ on the Argand Plane provided. 2
- (ii) Let the root with the smallest positive argument be α .
- (iii) Write the complex number that is in the fourth quadrant in terms of α . 1
- (iv) Write $\overline{\alpha^{-10}}$ in modulus argument form. 1

(c) Sketch the following loci and the answer sheet provided show all important features:

(i) $|z - 3 - 2i| = |z + 3|$ 2

(ii) $-\frac{\pi}{4} < \arg(z) < \frac{\pi}{4}$ and $\operatorname{Re}(z) \leq 2$ 2

(iii) $0 \leq \arg(z - 5 + 2i) \leq \frac{\pi}{2}$ and $|z - 5 + 2i| \leq 1$ 3

(iv) $|z - 3 + 4i| = 2$ and determine the maximum and minimum values for $|z|$ 3

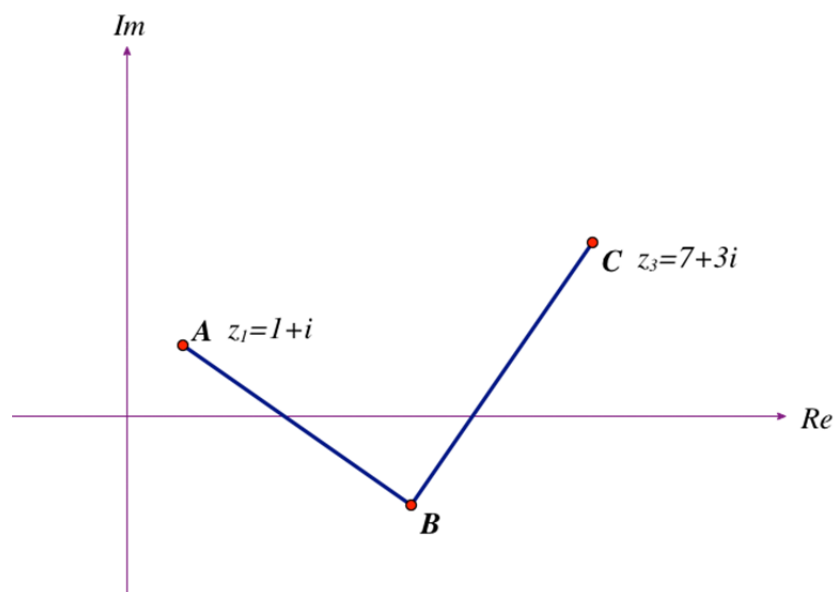
Question 12: (10 marks) Free Response

Marks

(a) Show that for any non zero complex number z 2
 $\arg\left(\frac{z}{\bar{z}}\right) = 2 \arg(z)$

(b) (i) Find the square roots of $5 - 12i$. 2
 (ii) Hence solve the equation $z^2 - (1 - 4i)z - (5 - i) = 0$, expressing your answers in the form $a + ib$. 2

(c)



The points A and C represent the complex numbers $z_1 = 1 + i$ and $z_3 = 7 + 3i$ and B represents the complex number z_2 . Find the complex number z_2 in the form $a + ib$ where a and b are real, such that $\triangle ABC$ is isosceles and right-angled at B.

4

\Student Number: _____ Mathematics Class: _____

Multiple choice answer sheet – Use pencil to completely colour the circle representing your answer.

1. (A) (B) (C) (D)

6. (A) (B) (C) (D)

2. (A) (B) (C) (D)

7. (A) (B) (C) (D)

3. (A) (B) (C) (D)

8. (A) (B) (C) (D)

4. (A) (B) (C) (D)

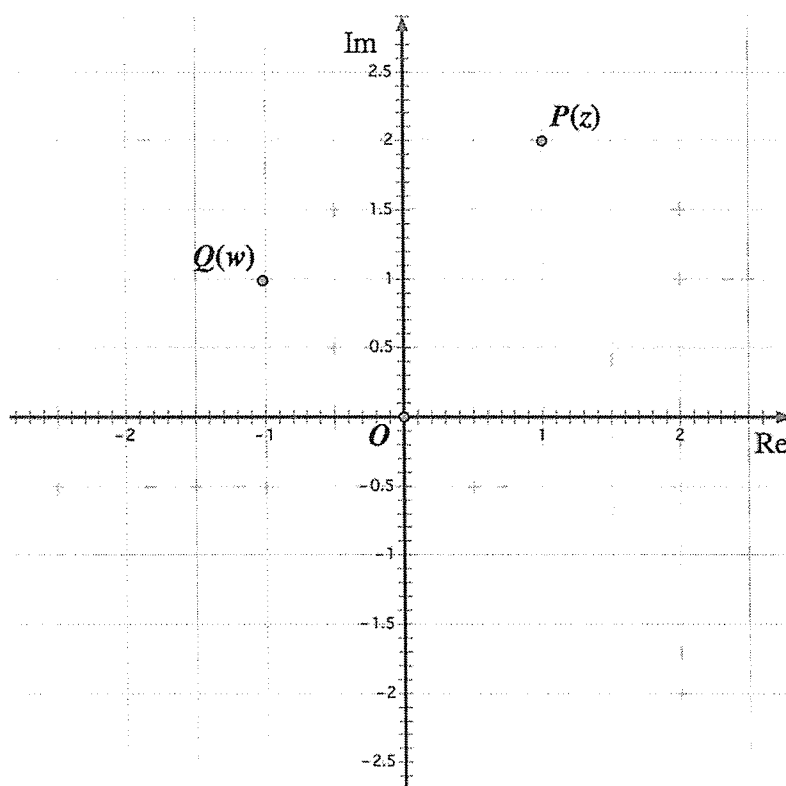
9. (A) (B) (C) (D)

5. (A) (B) (C) (D)

10. (A) (B) (C) (D)

Constrained Answer Sheet – Write your answer in the space provided.

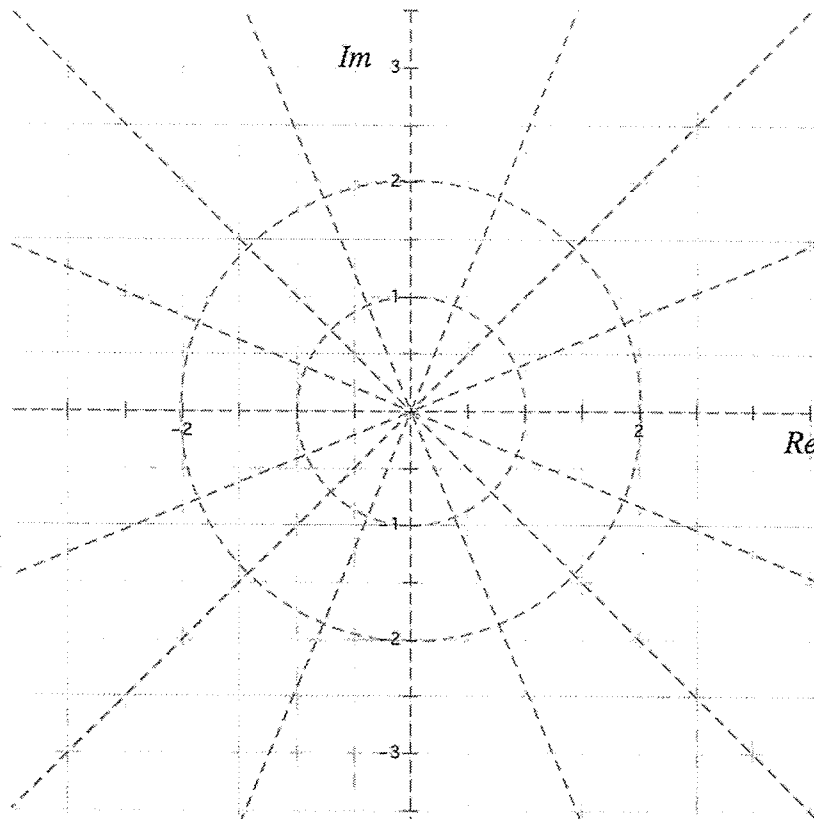
11 (a)



Plot the points that represent

- (i) $A(\bar{z})$
- (ii) $B(-\omega)$
- (iii) $C\left(\frac{1}{\omega}\right)$
- (iv) $D(iz)$
- (v) $E(\omega - z)$

(b) $z^8 - 1 = 0$



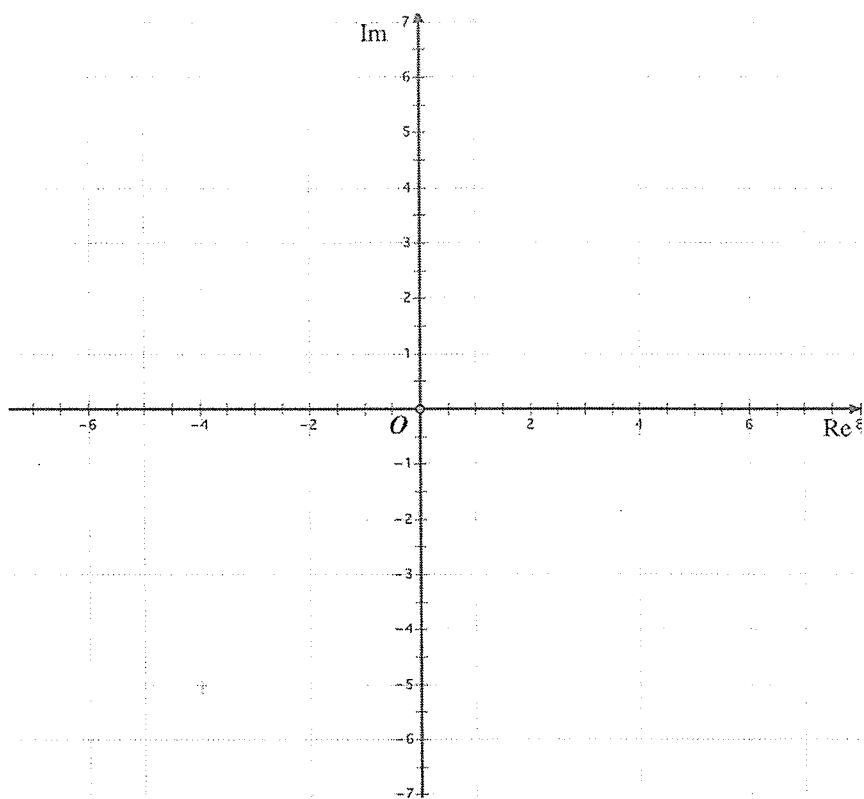
(iii) Write the complex number that is in the fourth quadrant in terms of α

(iv) Write $\overline{\alpha^{-10}}$ in modulus argument form.

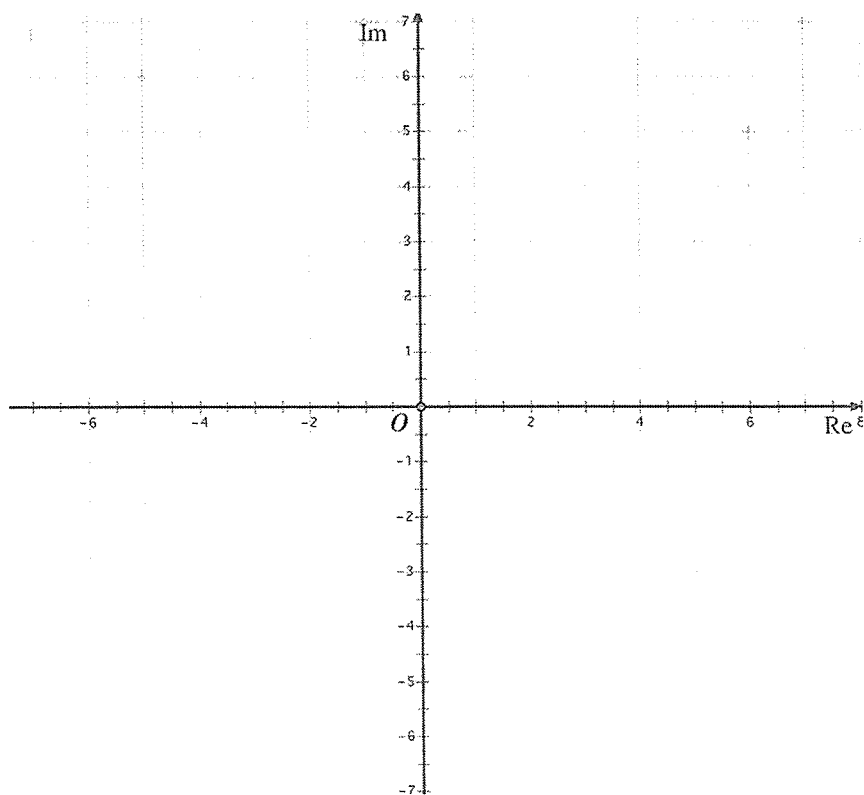
Student Number: _____ Mathematics Class: _____

11.(c)

(i) $|z - 3 - 2i| = |z + 3 + 2i|$

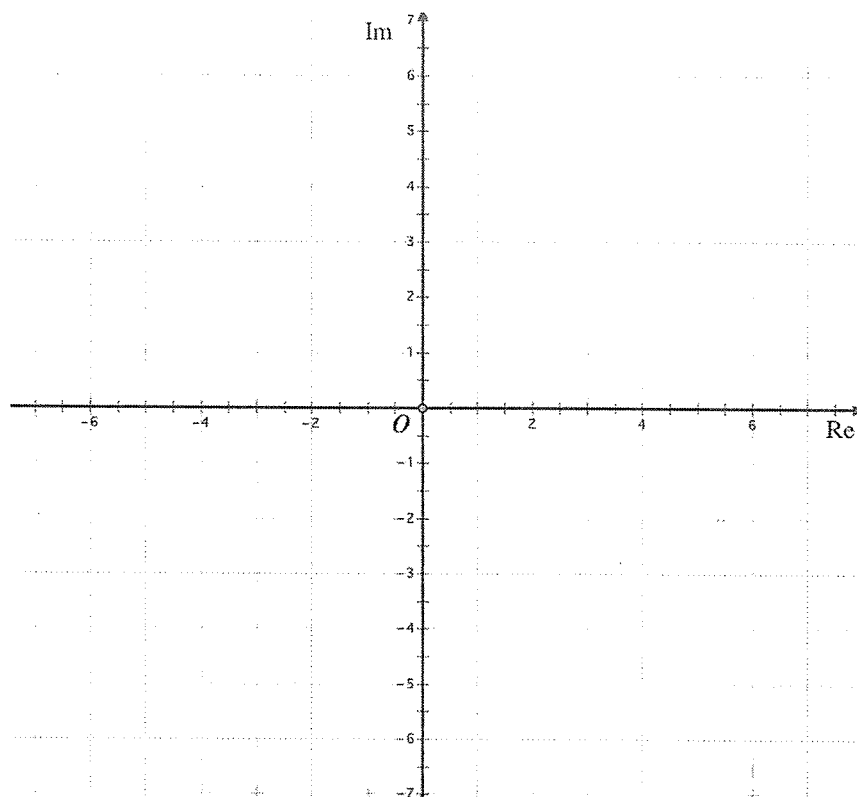


(ii) $-\frac{\pi}{4} < \arg(z) < \frac{\pi}{4}$ and $\operatorname{Re}(z) \leq 2$

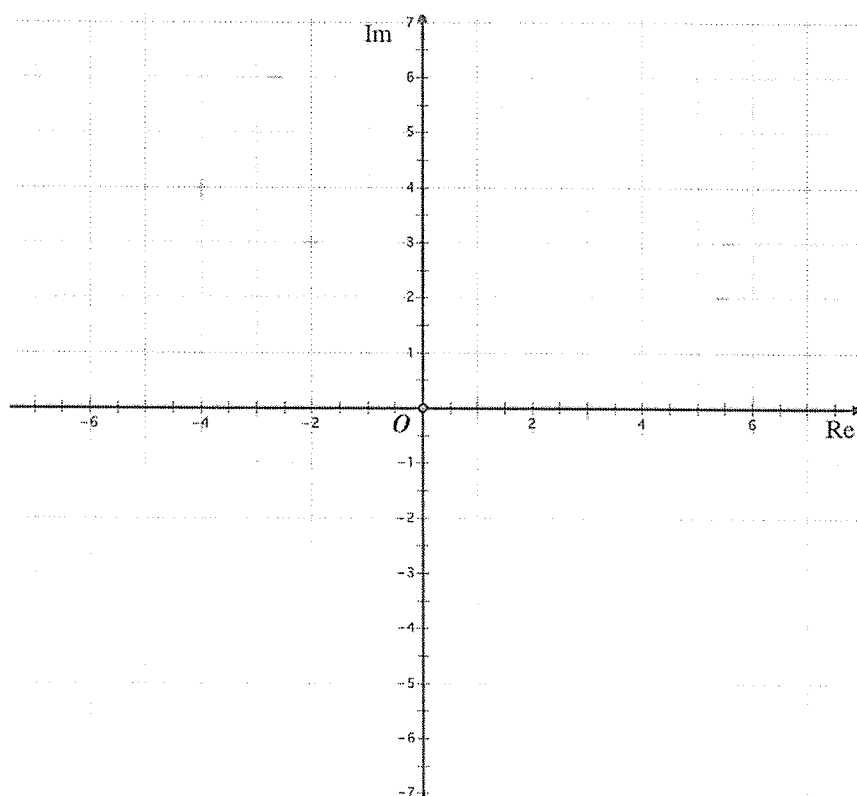


11.(c)

(iii) $0 \leq \arg(z - 2 + 2i) \leq \frac{\pi}{2}$ and $|z - 2 + 2i| \leq 2$



iv) $|z - 3 + 4i| = 2$



Maximum $|z|$ _____

Minimum $|z|$ _____

\Student Number: _____ Mathematics Class: _____

Multiple choice answer sheet – Use pencil to completely colour the circle representing your answer.

1. ☐ A ☒ B ☐ C ☐ D

2. ☐ A ☐ B ☐ C ☒ D

3. ☐ A ☒ B ☐ C ☐ D

4. ☐ A ☐ B ☒ C ☐ D

5. ☐ A ☐ B ☐ C ☒ D

6. ☒ A ☐ B ☐ C ☐ D

7. ☐ A ☐ B ☐ C ☒ D

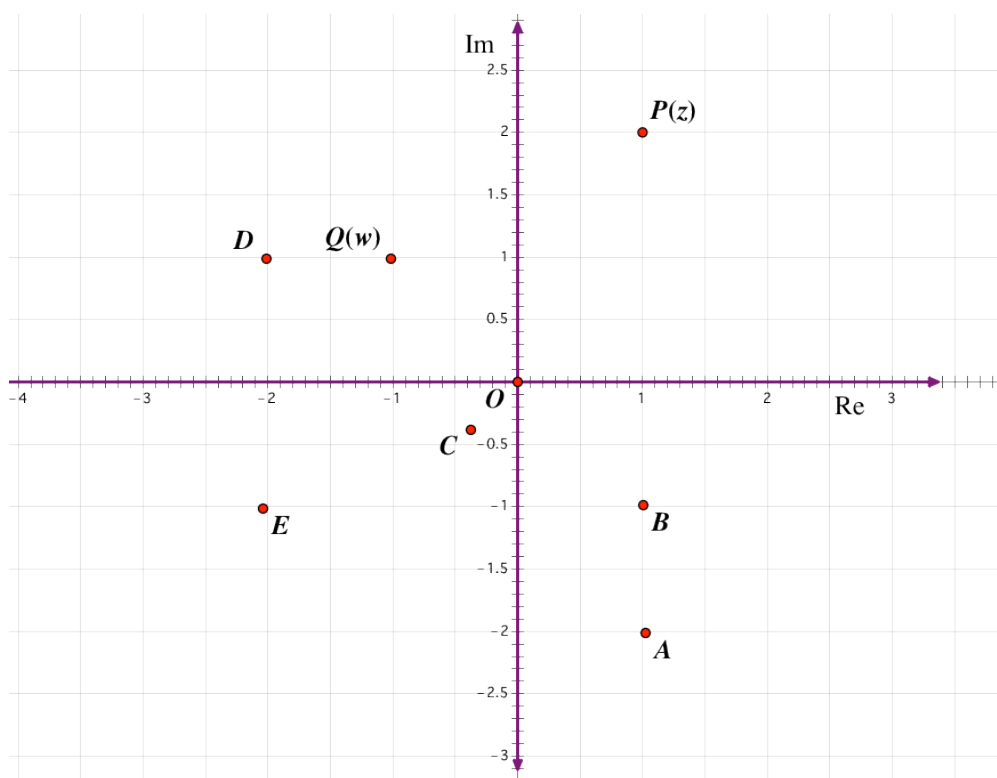
8. ☐ A ☐ B ☒ C ☐ D

9. ☐ A ☐ B ☒ C ☐ D

10. ☐ A ☒ B ☐ C ☐ D

Constrained Answer Sheet – Write your answer in the space provided.

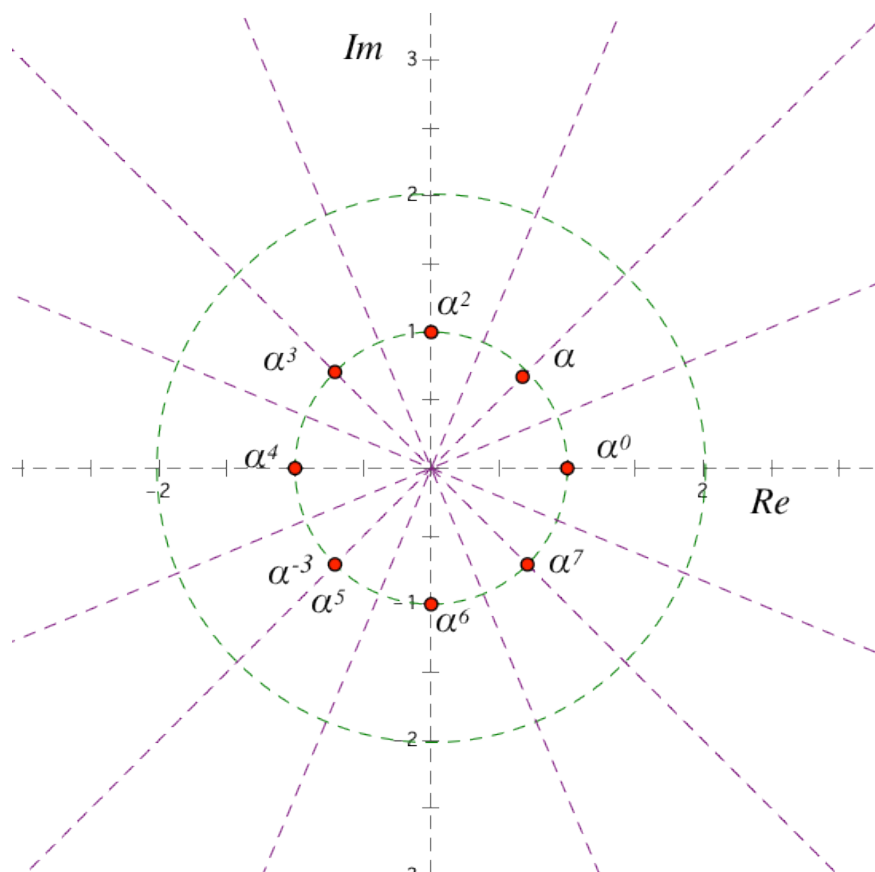
11 (a)



Plot the points that represent

- (i) $A(\bar{z})$
- (ii) $B(-\omega)$
- (iii) $C\left(\frac{1}{\omega}\right)$
- (iv) $D(iz)$
- (v) $E(\omega - z)$

(b) $z^8 - 1 = 0$



- (iii) If α is the complex root in quadrant 1, write the complex root in quadrant 4 as a power of α .

α^7 or α^{-1}

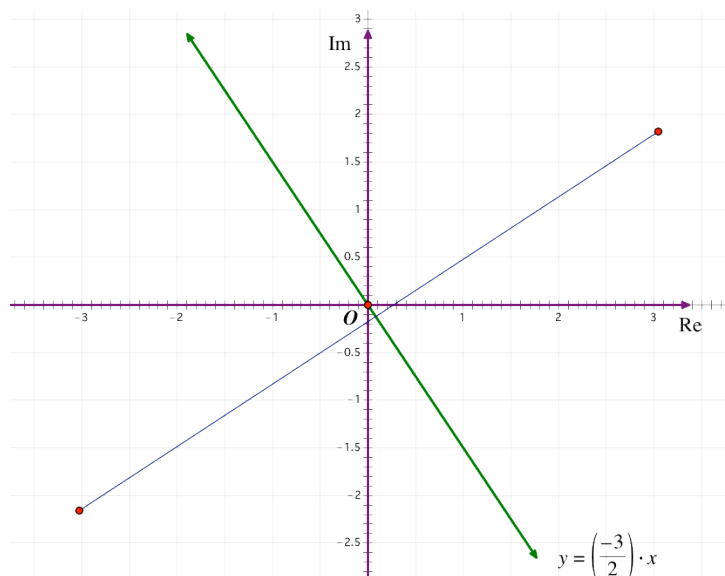
- (iv) Write $\overline{\alpha^{-10}}$ in modulus argument form.

$$\overline{\alpha^{-10}} = \overline{\alpha^6} = \alpha^2 = \cos\left(\frac{\pi}{2}\right) + i \sin\left(\frac{\pi}{2}\right)$$

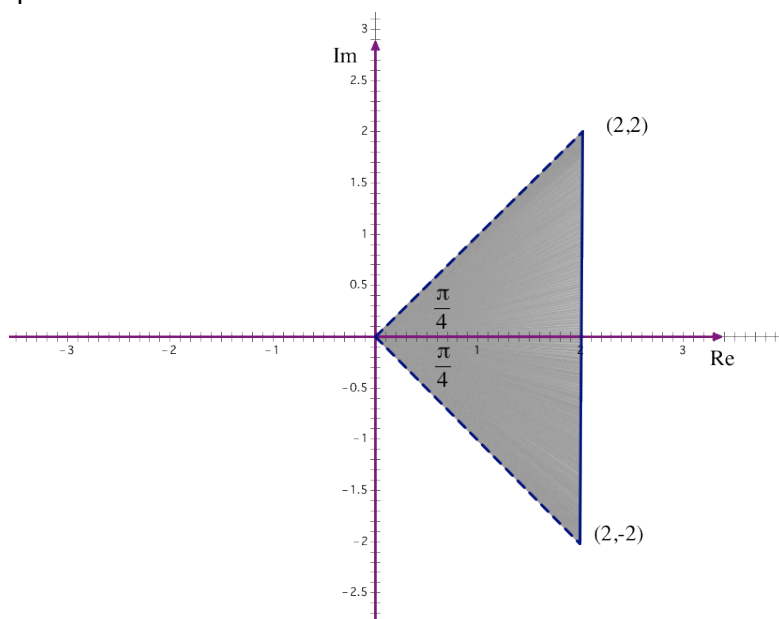
Student Number: _____ Mathematics Class: _____

11.(c)

(i) $|z - 3 - 2i| = |z + 3 + 2i|$

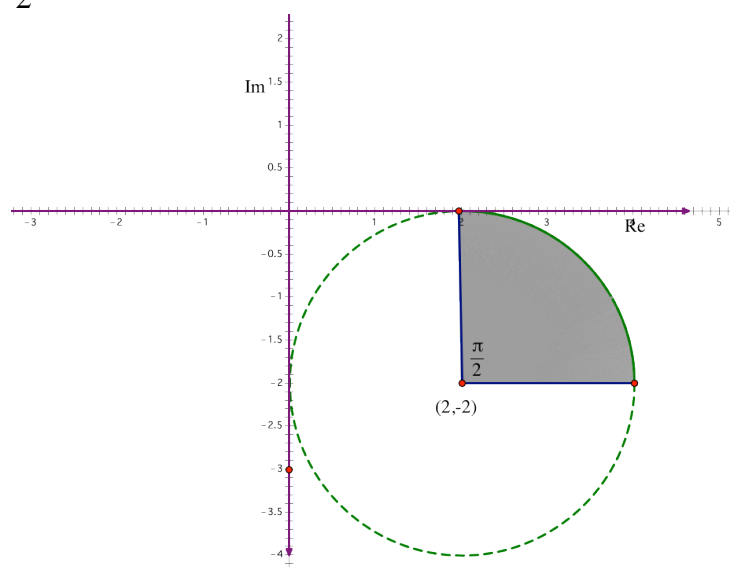


(ii) $-\frac{\pi}{4} < \arg(z) < \frac{\pi}{4}$ and $\operatorname{Re}(z) \leq 2$

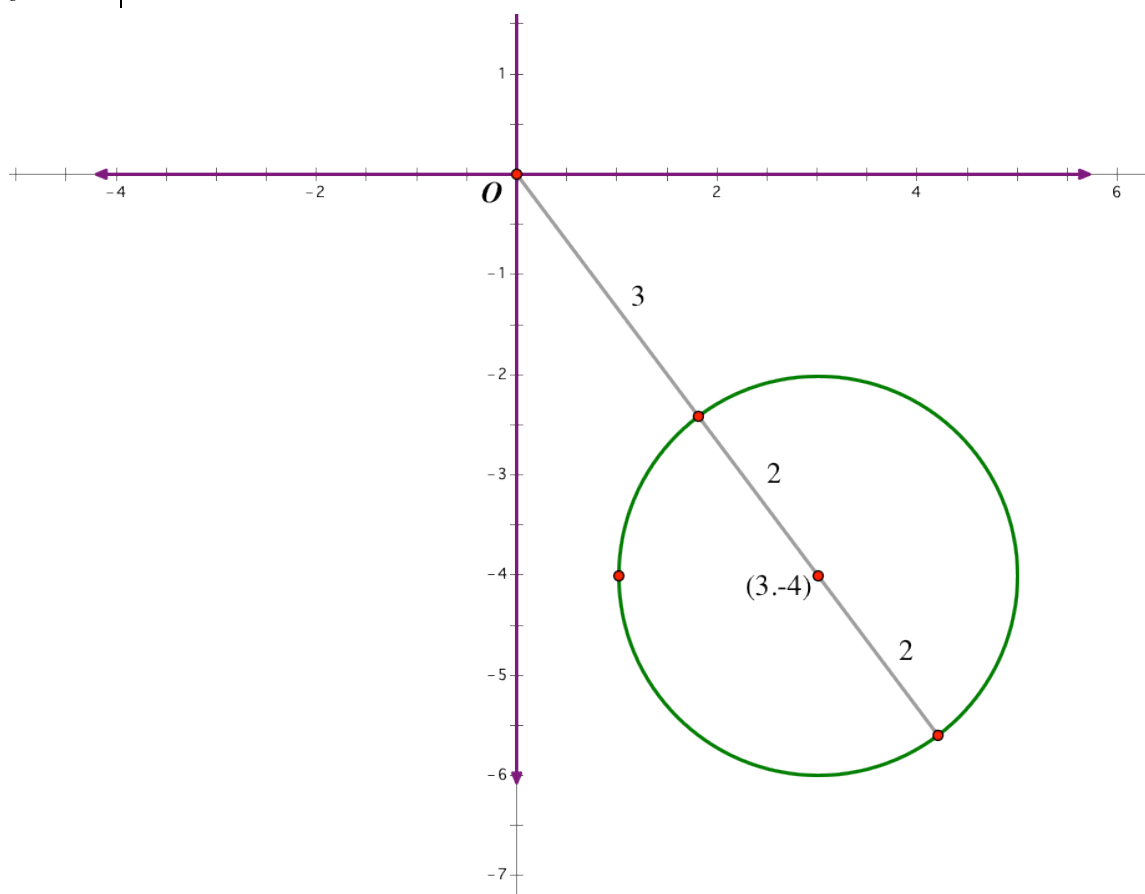


11.(c)

(iii) $0 \leq \arg(z - 2 + 2i) \leq \frac{\pi}{2}$ and $|z - 2 + 2i| \leq 2$



iv) $|z - 3 + 4i| = 2$



Maximum $|z|$ _____ = 7

Minimum $|z|$ _____ = 3

(a) Show that for any non zero complex number $\arg\left(\frac{z}{\bar{z}}\right) = 2\arg(z)$

2

Let $z = r\text{cis}\theta$ and so $\bar{z} = r\text{cis}(-\theta)$

$$\frac{z}{\bar{z}} = \frac{r\text{cis}\theta}{r\text{cis}(-\theta)} = \text{cis}(2\theta)$$

$$\therefore \arg\left(\frac{z}{\bar{z}}\right) = 2\theta = 2\arg z$$

(b)

(i) Find the square roots of $5 - 12i$.

$$\begin{aligned} 5 - 12i &= (x + yi)^2 \\ &= x^2 - y^2 + 2xyi \end{aligned}$$

$$\begin{aligned} 5 &= x^2 - y^2 \\ -6 &= xy \end{aligned}$$

$$\therefore x = -3 \text{ and } y = 2$$

$$\therefore x = 3 \text{ and } y = -2$$

Square roots of $5 - 12i$ are $-3 + 2i$ and $3 - 2i$

(ii) Hence solve the equation $z^2 - (1 - 4i)z - (5 - i) = 0$, expressing your answers in the form $a + ib$.

2

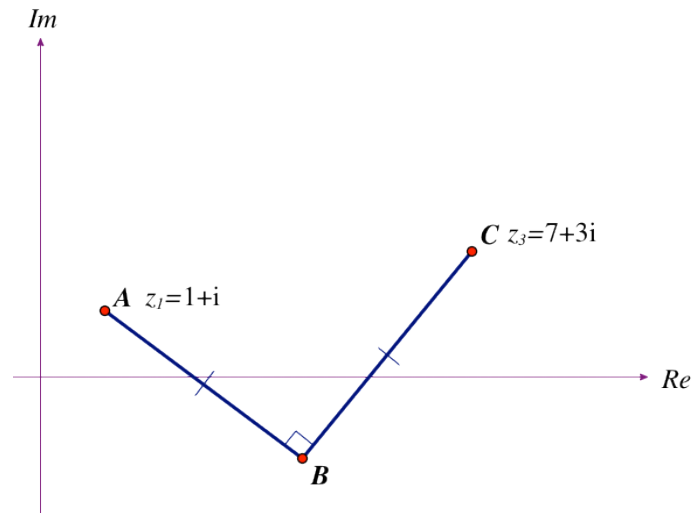
$$z = \frac{1 - 4i \pm \sqrt{(-1 + 4i)^2 - 4 \times 1 \times -(5 - i)}}{2}$$

$$z = \frac{1 - 4i \pm \sqrt{5 - 12i}}{2}$$

$$z = \frac{1 - 4i \pm (3 - 2i)}{2}$$

$$z = \frac{4 - 6i}{2} = 2 - 3i \quad \text{and} \quad z = \frac{-2 - 2i}{2} = -1 - i$$

(c)



The points A and C represent the complex numbers $z_1 = 1 + i$ and $z_3 = 7 + 3i$ and B represents the complex number z_2 . Find the complex number z_2 in the form $a + ib$ where a and b are real, such that $\triangle ABC$ is isosceles and right-angled at B .

4

$$\begin{aligned}\vec{BA} &= z_1 - z_2 \\ &= 1 + i - a - bi \\ &= 1 - a + i(1 - b)\end{aligned}$$

$$\begin{aligned}\vec{BC} &= z_3 - z_2 \\ &= 7 + 3i - a - bi \\ &= 7 - a + i(3 - b)\end{aligned}$$

$$i \times \vec{BC} = \vec{BA}$$

$$i \times (7 - a + i(3 - b)) = 1 - a + i(1 - b)$$

$$i(7 - a) - (3 - b) = 1 - a + i(1 - b)$$

By equating real and imaginary parts

$$-(3 - b) = 1 - a$$

$$b = 4 - a$$

and

$$(7 - a) = (1 - b)$$

$$7 - a = 1 - (4 - a)$$

$$7 - a = -3 + a$$

$$10 = 2a$$

$$a = 5$$

$$b = 4 - a$$

$$b = 4 - 5$$

$$b = -1$$

$$\therefore z_2 = 5 - i$$